Time Crystals

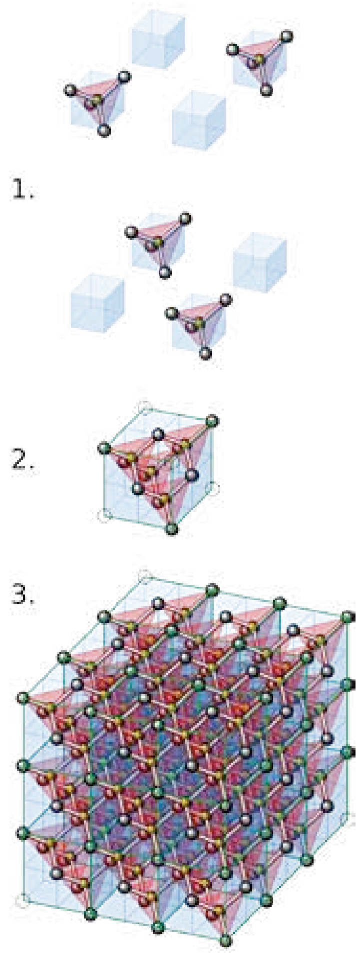
Completely warping our idea of time and energy

Can you run to and fro without expending any energy? Well, Time Crystals can.

Ronak Gupta
feedback@digit.in

Fancy theories are commonplace in physics, but when a Nobel Laureate comes up with one, they are taken seriously. In 2012, Nobel Laureate Frank Wilczek hypothesised the existence of a type of material called ‘time crystals’. Initially, they seemed to defy the laws of physics. Recently, these mysterious time crystals have been constructed in a laboratory – giving creedence to Wilczek’s bizarre proposition. In this article, we breakdown time crystals – why are they weird, how were they created and what can they mean for the future of physics.

When it comes to the laws of physics, symmetry is sacrosanct. Laws of physics are ‘spatially’ symmetric, as in no position in space is more favoured than the other. What if these symmetries are broken? Indeed, spatial symmetry breaking does happen often in nature. The gravitational force that binds the earth in orbit around the sun is spherically symmetric – not having any directional dependency, depending only on the distance between the two bodies. The motion resulting from this force however breaks symmetry by executing elliptical orbits which aren’t symmetric about its axes, unlike a sphere. This is an example of symmetry breaking in an ‘excited’ state of energy. More interesting is the case of symmetry breaking in the ground (lowest energy) state of the system, and this happens in the case of crystals. Crystals, like diamond, have lattice structures which repeat in space. They show discrete symmetry and periodicity which violates the above equivalence of space principle accorded by spatial symmetry. Now, the laws of physics are also symmetric in time, which disallow stable objects to change form over time. Thanks to Einstein and co-workers, we are used to unifying space and time into a single four-dimensional space. So, if spatial symmetry can be broken, can there be cases where temporal symmetry breaks?



Visualisation of a diamonds lattice. Repeating structures are clearly visible

Time crystals do exactly that. The molecular structure of a crystal forms a pattern in space by repeating itself over regular intervals. The same way, time crystals ‘repeat’ structures over time intervals. What does time symmetry breaking mean? How does it manifest itself? Think of a body attached to a perfect spring dangling in frictionless air. If we perturb this system by slightly pulling the string, the body will execute a periodic motion in time, returning to its original position regularly. If drag was there, this periodic motion would die down and the body would come to rest. This periodicity in time is nothing special and plenty of systems execute such motion. Periodic motion is not symmetric in time, but such a motion does not count as time symmetry breaking. This is because most periodic motions are associated with some inherent energy. When we perturbed the spring by tugging at, we gave it some potential energy, which keeps getting converted into energy of motion and vice versa, keeping the system in a state of constant energy. Would the block do the same motion without the initial tug? Can a system have time periodicity in a state of zero or lowest energy? The answer is yes. When there is a time repeating pattern in the system at ground state – it indicates breakdown of time symmetry. Time crystals keep oscillating in time and here’s the clincher. The oscillation occurs without any energy.

Imagine a collection of ions arranged in the form of a ring. If we cool them down to a temperature close to zero, the